

Neural Network Zoo: Why the Wolf Spider Represents a GNN

Graph Neural Networks • Animal Metaphor • Message Passing

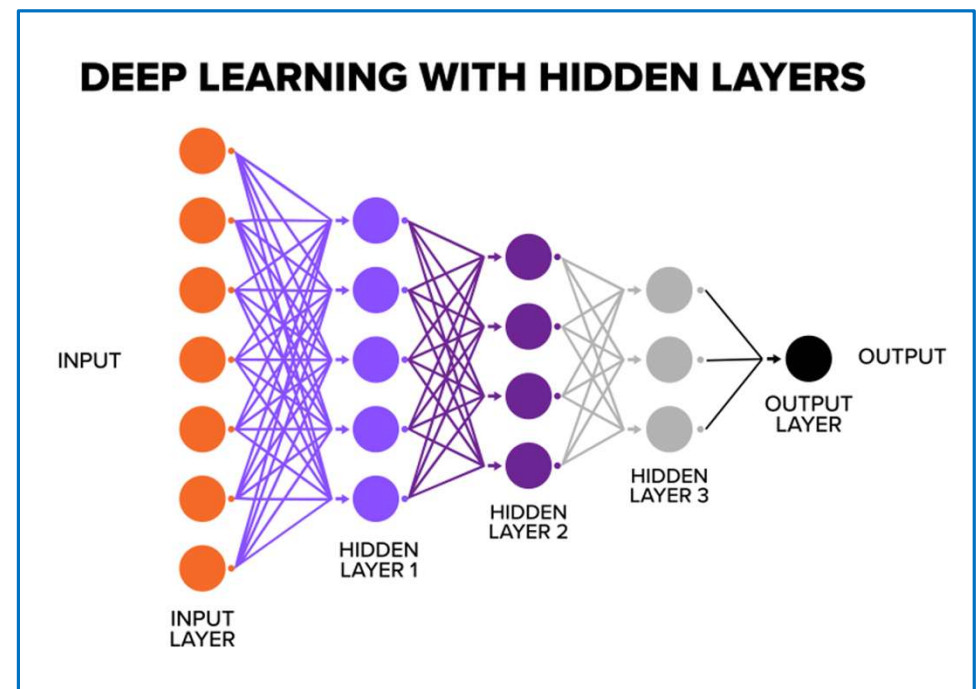
What is Deep Learning and Neural Networks?

What Is Deep Learning?

It is a branch of artificial intelligence that uses multi-layer neural networks to learn complex patterns from data. For example, a network can learn to recognize features in an image. This ability allows Deep Learning to be used in different areas of artificial intelligence.

Important concepts:

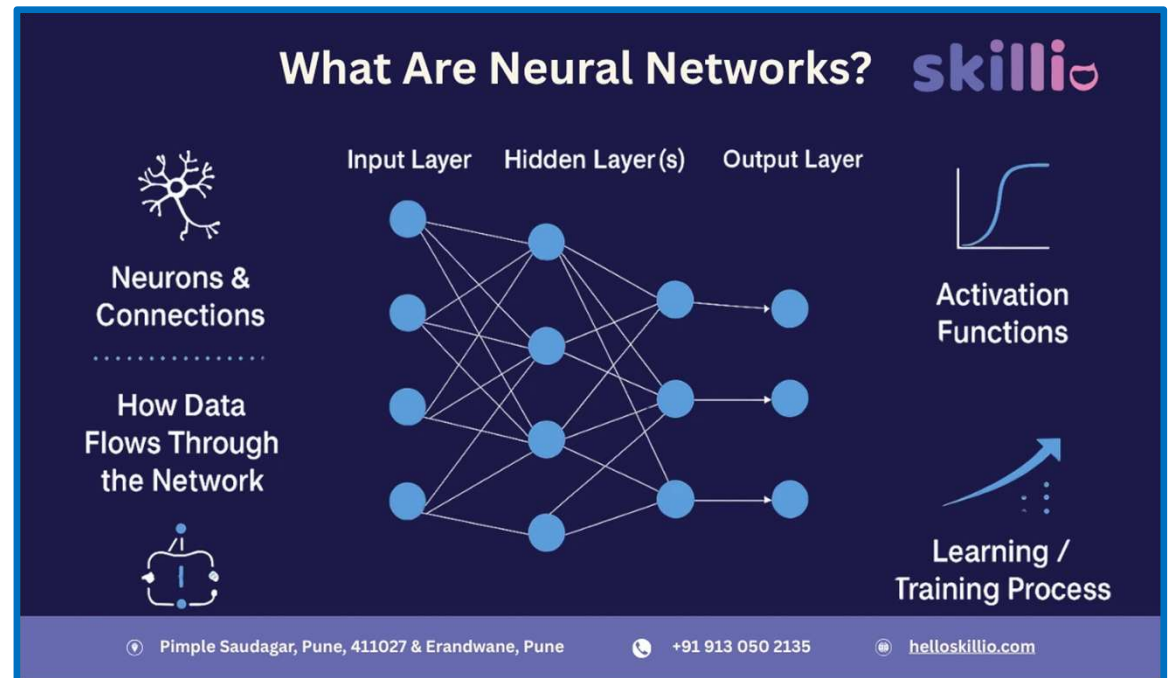
- Uses artificial neural networks.
- Features multiple processing layers.
- Learn complex patterns from data.
- Used for images, audio, text, and video.



What Is a Neural Network?

Neural networks are made of artificial neurons connected to each other, but each neuron receives information, processes it, and produces an output.

Neurons are organized into layers: an input layer, hidden layers, and an output layer. This organization allows the network to process information and learn patterns.



Neurons Organization:

Input Layer → Hidden Layers → Output Layer

How Does a Neural Network Learn?

Neural networks learn through a training process because it receives data and produces predictions. Then, it compares that prediction with the expected result and adjusts its weights to reduce the error. When repeating this process with many examples, the network learns patterns that help make better predictions.

Process:

Data → Prediction → Comparison → Weight Adjustment → Better Prediction

The Neural Network Zoo:

Just as animals have unique capabilities for survival, different neural network architectures specialize in distinct tasks. Our team explores this concept through our chosen model and animal analogy.

Where Does a GNN Fit?

Deep learning expands neural networks by stacking many layers.

Different architectures are designed for different kinds of data:

- Images → Convolutional Neural Networks (CNNs)

- Sequences → Recurrent/Transformer-based models

- Graphs → Graph Neural Networks (GNNs)

A GNN is specialized for data where relationships between things are important.

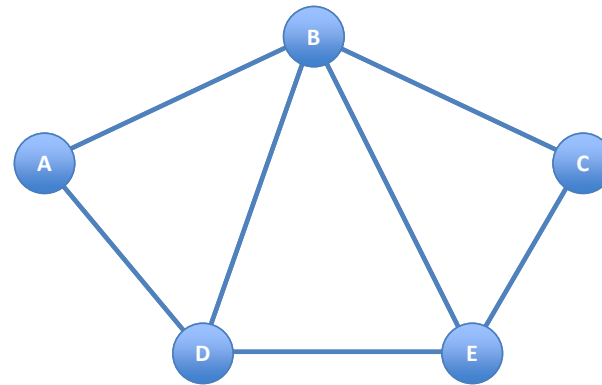
The Basic Idea of a Graph

A graph is a way to represent entities and the relationships between them.

Nodes = the entities or objects.

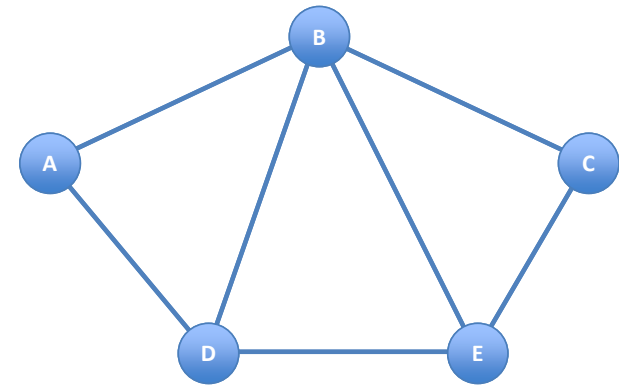
Edges = the relationships or connections.

Examples: people connected by friendships, cities connected by roads, or atoms connected by chemical bonds.



How Does a GNN Learn? — Message Passing

1. Each node receives information from its neighboring nodes.
2. It combines the neighbors' information with its own features.
3. It updates its internal representation.
4. After several rounds, each node captures information about its local neighborhood and the larger graph.



Why Are GNNs Useful?

GNNs are useful when relationships are as important as the individual data points.

Molecules → predict properties and behavior

Fraud detection → identify suspicious relationship patterns

Social networks → analyze communities and connections

Recommendation systems → connect users, products, and interactions

Traffic and route optimization → model roads, intersections, and routes

Choosing the Animal: Three Candidates

Our challenge: choose an animal that communicates the most important GNN ideas.

 Wolf — teamwork and communication

 Spider — webs, connections, and signals

  Wolf spider — combines the strengths of both metaphors

Key question: Which animal best represents connectivity + information flow?

Candidate #1 — Wolf

Pros

- Wolves operate in packs, loosely resembling nodes interacting.
- They communicate and coordinate, suggesting message passing.

Cons

- A wolf pack is not a structured network of nodes and edges.
- Pack relationships are more social/hierarchical than graph-like.
- The wolf does not naturally communicate network topology.

Conclusion: The wolf represents teamwork, but it is the weakest fit for GNN structure.

GNN WOLF

GRAPH NEURAL NETWORK NETWORK STRUCTURE

*Stronger
Through
Connections*

A **Graph Neural Network (GNN)** works with data that can be represented as a graph. A graph is made of **nodes** and **edges**, where the connections between objects are important.



Nodes

- Represent individual objects or entities
- Example: each wolf in the pack is a node



Edges

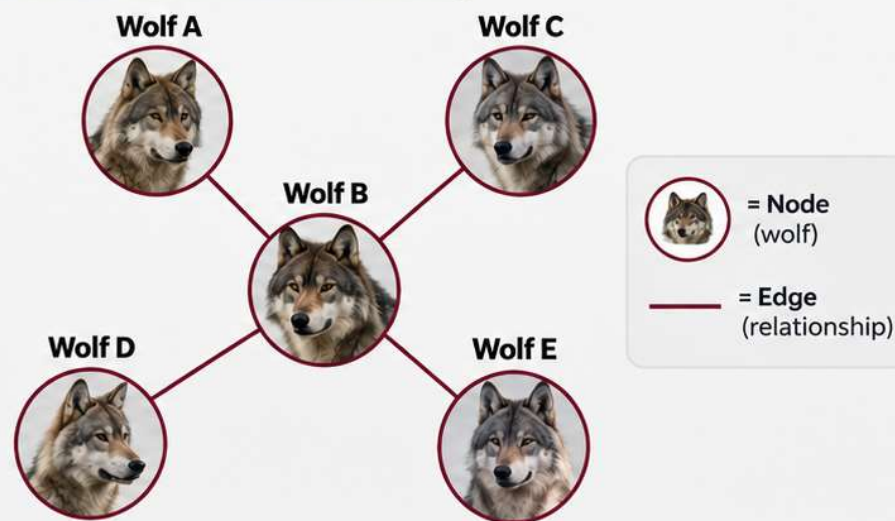
- Show the relationships or connections between nodes
- Example: wolves that interact with each other



Features

- Each node can have information (features)
- Edges can also have features describing the relationship

Example: Wolf Pack as a Graph



DIFFERENT INDIVIDUALS.
A STRONGER WHOLE.



GNN WOLF

GRAPH NEURAL NETWORK

Information Flows Through the Pack

*Together
We Learn
More*

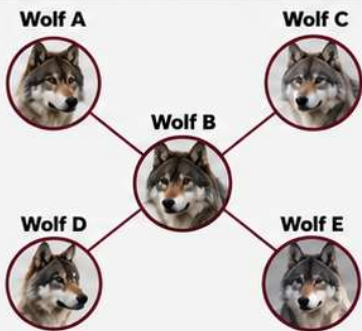
SAME
CONNECTIONS.
STRONGER
PREDICTIONS.

How Information Moves in a GNN

A GNN uses **message passing**. Each node receives information from its neighbors, combines it with its own information, and updates its representation. This process happens in **multiple layers**.

1. Input Graph

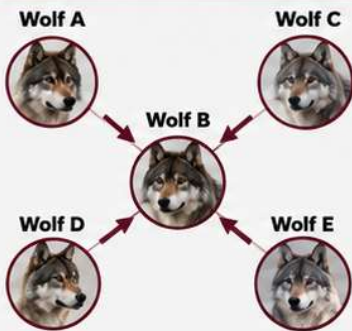
Nodes with initial features



Each wolf (node) has its own information (features).

2. GNN Layer 1

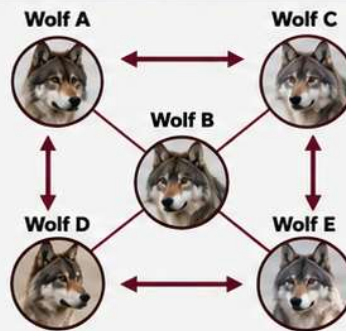
Receive information from neighbors



Each node receives information from its direct neighbors.

3. GNN Layer 2

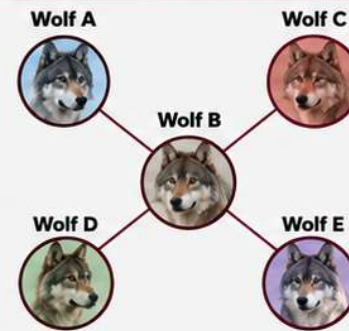
Information travels farther



With another layer, information can travel to nodes that are farther away in the graph.

4. Output

Updated node representations



After several layers, each node has an updated representation that can be used for different tasks (classification, prediction, etc.).

Key Points



- 1 Nodes start with their own features.
- 2 Each layer allows nodes to receive and combine information from neighbors.
- 3 More layers help the network learn about the larger graph.
- 4 The final output is used for tasks like node classification, link prediction, or graph classification.

*A stronger
pack through
shared knowledge.*





GNN WOLF

GRAPH NEURAL NETWORK

Different Wolves. Stronger Together.

*Connections
Create
Intelligence.*

What Can a GNN Predict?

After processing the graph, a GNN can be used for different tasks, depending on the problem.



Node Classification

Predicting what type of node something is (example: identify the role of a wolf).



Link Prediction

Predicting if two nodes should have a connection (example: will two wolves become part of the same pack?).



Graph Classification

Predicting something about the entire graph (example: classify the behavior of the whole pack).

Why the Wolf Represents a GNN?

Wolves live in packs and depend on each other. They share information, work together, and make decisions based on the other members of the pack. This is very similar to how a GNN works.

Wolf Pack		GNN Concept
Individual wolves	→	Nodes
Relationships in the pack	→	Edges
Sharing information	→	Message passing
Pack decisions	→	Learned representation
Stronger together	→	Better predictions



*A pack is more
than individual
wolves. It is the
connections between
them that make
them stronger.*

Key Takeaways

- 1 GNNs learn from the relationships between nodes.
- 2 Message passing allows information to flow through the graph.
- 3 More connections lead to better understanding and predictions.

*More Connections.
A Smarter Future.*

Candidate #2 — Spider

Pros

A spider web is a literal network of connected points and pathways.

The web visually resembles nodes and edges.

A spider can detect vibrations traveling through the web — a useful analogy for information flow.

Cons

The basic spider metaphor can feel generic.

Not all spiders build capture webs.

Conclusion: The spider is much closer to GNNs, especially for connectivity and message passing.

Candidate #3 — Wolf Spider: The Final Choice

The wolf spider combines the strengths of the two metaphors.

 Wolf → individual awareness and active interaction with the environment.

 Spider → sensing signals and navigating a connected environment.

The hybrid name mirrors the hybrid nature of a GNN: node features + relational structure.

Important clarification: wolf spiders do not build capture webs. The “web” is conceptual — the GNN’s network is a structure of nodes and edges.

The metaphor focuses on information propagation, not literal web construction.

Why the Wolf Spider Wins

Nodes → sensory points / individual processing units

Edges → pathways through which signals or information can be related

Message passing → information from the surrounding connected structure influences a node

Graph structure → the interconnected environment being analyzed

Most importantly: the analogy makes an abstract GNN easier to explain visually.

Final argument: The spider represents the graph; the wolf represents active interaction; the wolf spider brings the ideas together.

Why GNNs Matter in the Real World

GNNs are especially useful when the *relationships* between data points matter as much as the points themselves

Fraud Detection:

Transactions can be modeled as a graph of customers, accounts, merchants, and transfers. GNNs can detect suspicious relationship patterns that traditional methods may miss because they analyze the network structure itself.



Bank accounts + transactions → **suspicious network patterns** → **fraud alert**

Recommendation Systems:

Users, products, clicks, purchases, and ratings naturally form interconnected graphs. GNNs can learn from these relationships to improve recommendations rather than treating users or products as isolated records.



Users + products → **interaction patterns** → **recommendation**

Molecular & Biological Modeling:

Atoms can be represented as nodes and chemical bonds as edges, which makes molecular data inherently graph-structured. GNNs are widely used in biology and molecular analysis because the relationships among components are central to prediction.



Atoms + bonds → **molecular relationships** → **property prediction**

References and Contributions:

Slides 2-4: What are Neural Networks and Deep Learning (Cesar Noriega)

Slides 11 - 13: GNN Wolf (Luiz Paludo)

Slide 17: Why GNNs Matter in the Real World (Nichelle Graf)

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Wu, S., Sun, F., Zhang, W., Xie, X., & Cui, B. (2022). "Graph Neural Networks in Recommender Systems: A Survey." **ACM Computing Surveys**, 55(5).

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All other Slides: MaryAnn Mastri